

Integrated AI Systems Synthesis

Configurable Opportunity-to-Contract Intelligence & Assurance Framework

ICM Solutions reference implementation for
public-sector consulting decisions

Luis Muniz • Udacity AI Mastery Capstone • August 2026



01.

Mentor defense roadmap

15-minute story arc: problem → integrated system → evidence → governance → defense.

01

Problem

Public-sector
opportunity
decisions
under
uncertainty

02

System

Configurable
workflow with
evidence,
audit, and
human review

03

Synthesis

How Projects
1, 2, 4, and 6
operate
together

04

Evaluation

Scenario suite,
acceptance
assertions,
and
refinement

05

Governance

Safeguards,
nonbinding
recommendati
ons, residual
risks

06

Defense

Limitations,
industry value,
production
path

Defense theme: the framework supports professional judgment; it does not replace accountable human authority.

Industry context & problem definition

The problem is not finding opportunities; it is deciding responsibly under fragmented evidence.

ICM must evaluate public-sector opportunities using fragmented inputs: solicitation language, service fit, past demand, clauses, evidence, and specialist judgment.

AI is appropriate because the work is high-volume, cross-document, pattern-based, and evidence-intensive.

The constraint is decisive: public-sector legal, procurement, staffing, and reputational risk remain human responsibilities.

Core challenge: turn fragmented opportunity evidence into a defensible, auditable, nonbinding recommendation packet.



Human decision boundary: pursue, reject, defer, accept with conditions, or escalate remains an authorized human disposition.

03. Integrated AI system overview

A configurable opportunity-to-contract framework with visible handoffs and audit controls.

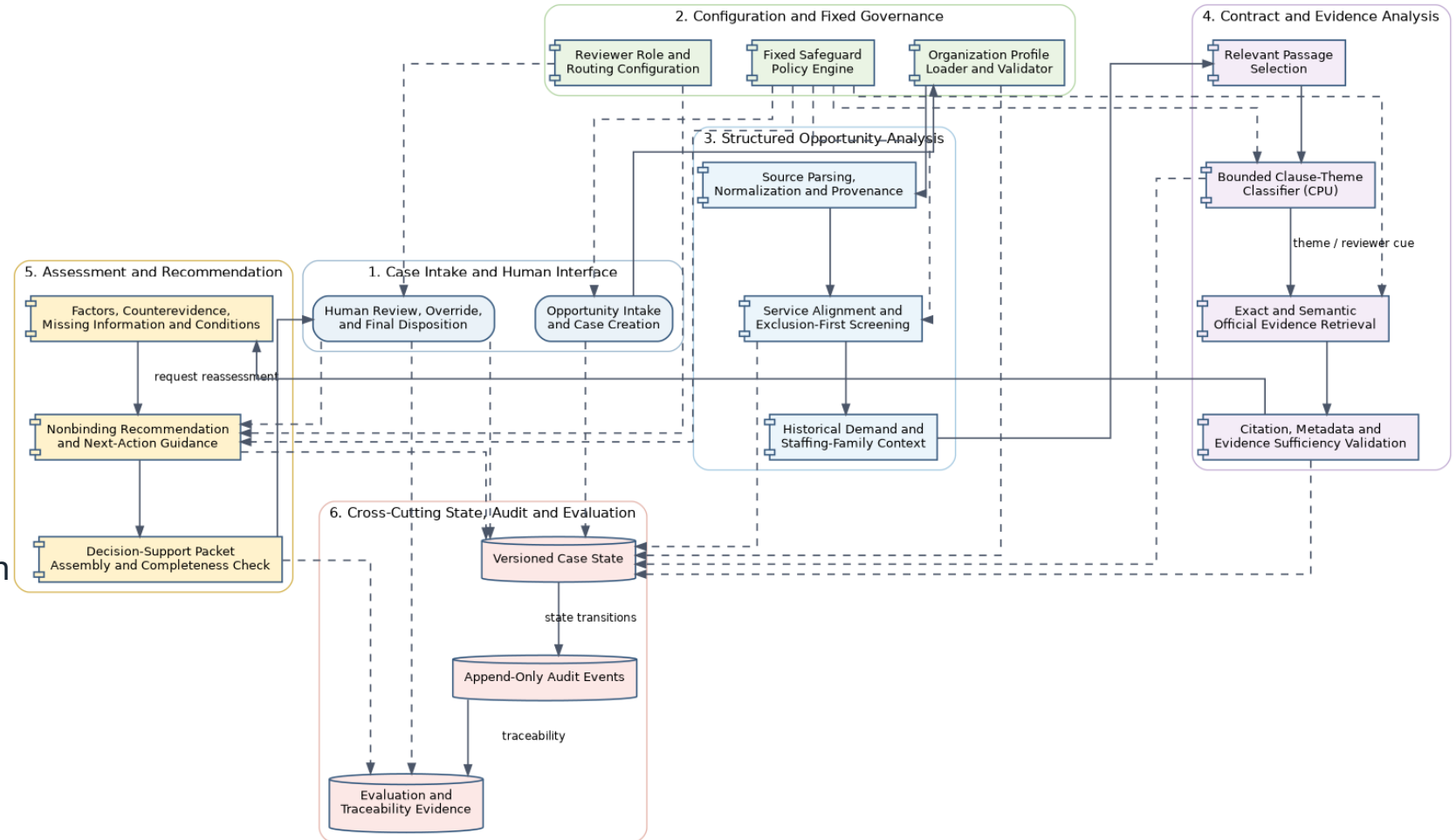
Inputs: opportunity metadata, source bytes, organization profile, selected clauses, and reviewer prompts.

Analysis: alignment context, bounded clause triage, evidence review, and recommendation logic.

Outputs: case state, audit trail, recommendation packet, limitations, and required reviewer role.

Boundary: nonbinding AI recommendation stays separate from final human disposition.

Artifact type: new integrative system/workflow, not a resubmission of any prior project.

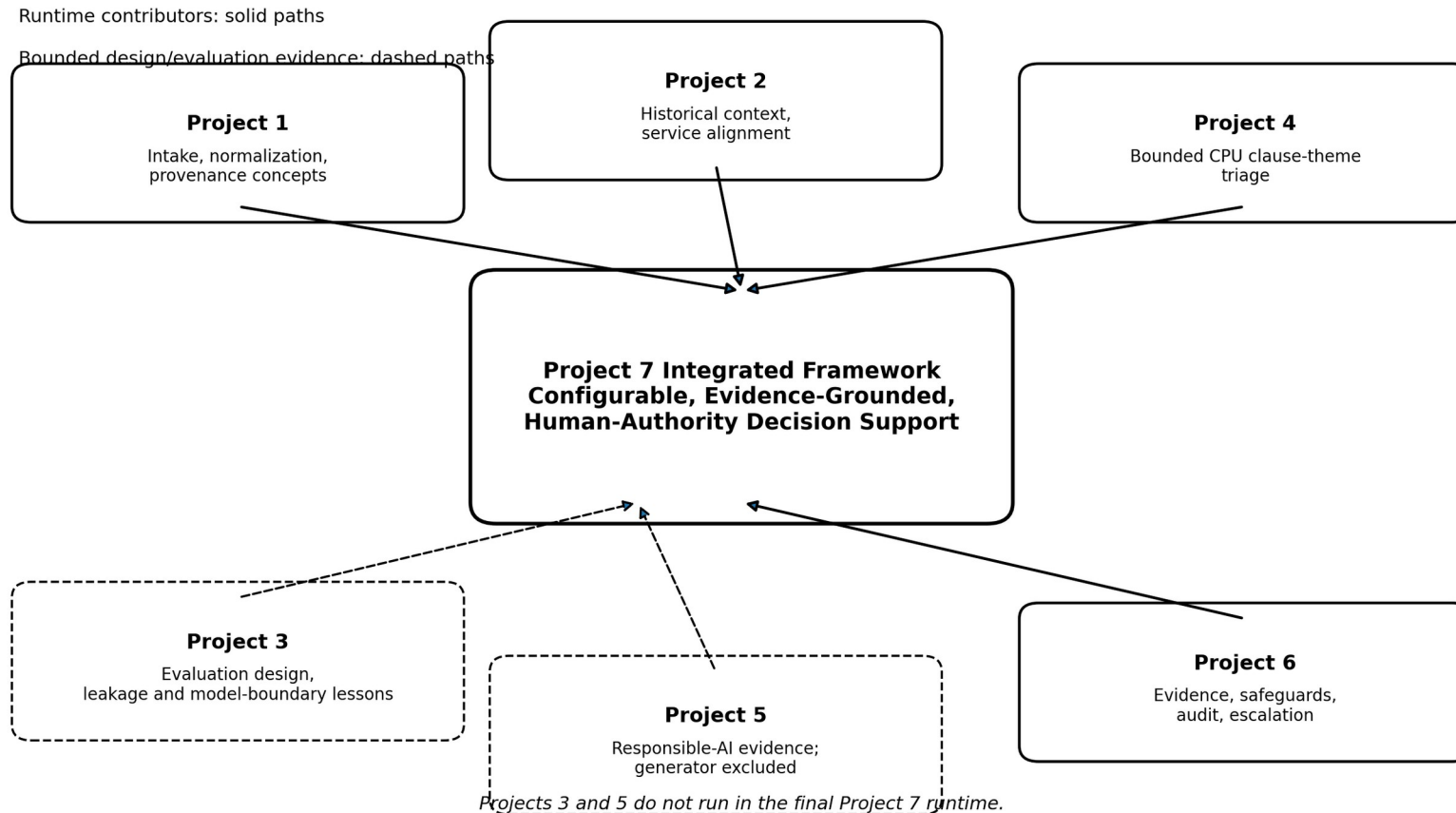


04.

Integration of prior capstone projects

The final system uses prior work intentionally—and excludes components that would overclaim.

Prior-Capstone Integration into the Final Project 7 System



Runtime contributors

Project 1: intake, normalization, provenance

Project 2: historical procurement context

Project 4: bounded clause-theme triage

Project 6: evidence, state, safeguards, escalation

Bounded design evidence

Project 3: target discipline / leakage control

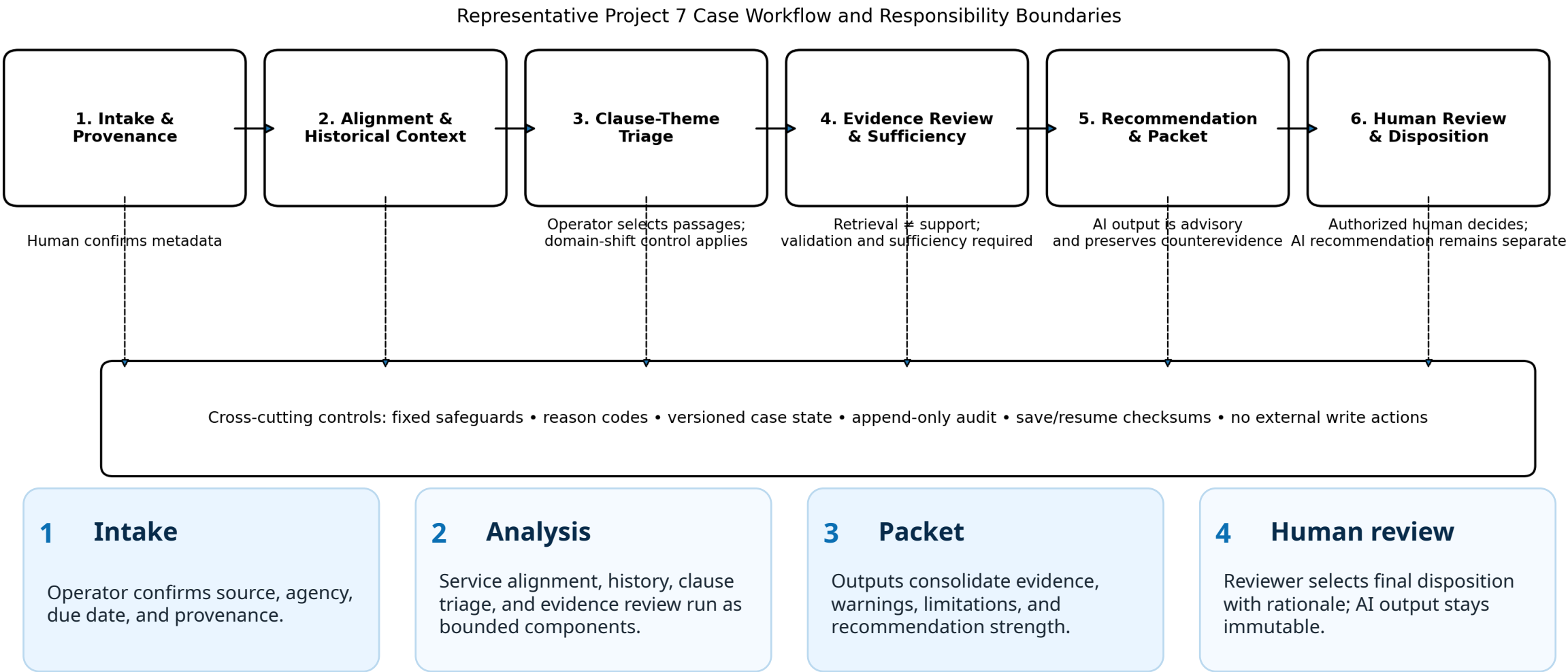
Project 5: why generative text is not authority

Synthesis means combining useful methods while preserving each component's boundary.

05.

Representative end-to-end scenario

EDD CalJOBS IV&V request used to demonstrate the workflow and responsibility boundaries.



Representative outcome: specialized review was escalated rather than overstating unsupported evidence.

06.

Recommendation packet and human authority

The AI recommendation is preserved as advice; the authorized disposition remains human-controlled.

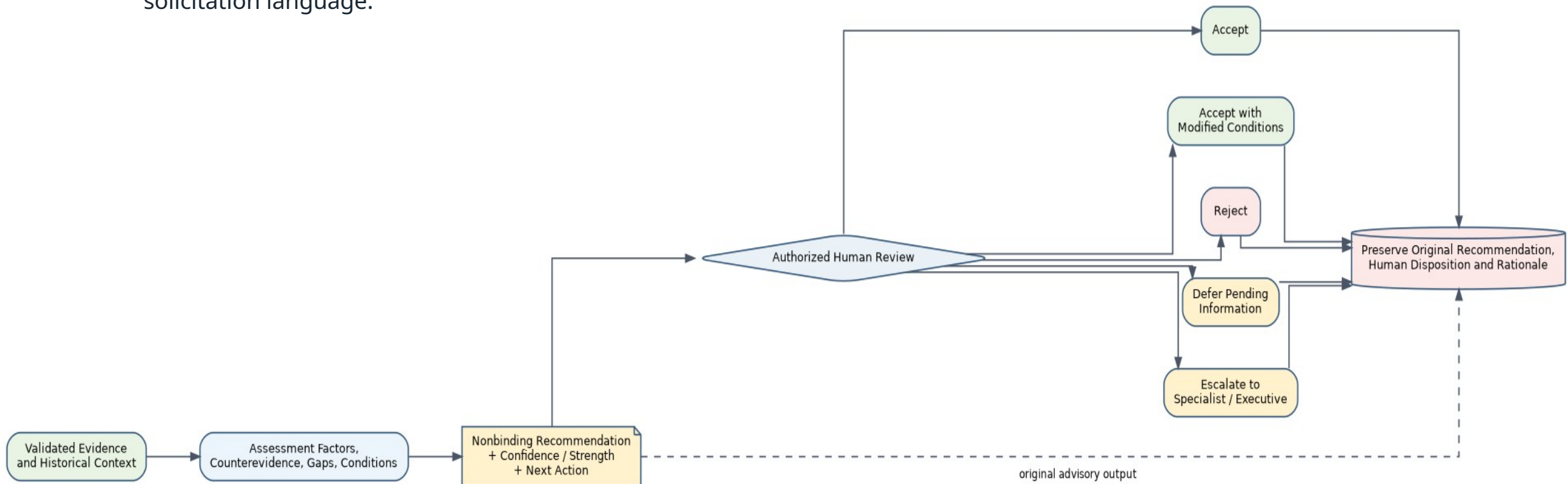
EDD case result: R-05 — Escalate: Specialized Review Required

Clause triage: assignment, insurance, and ownership warnings routed for specialist review.

Evidence sufficiency: retrieved FAR examples were not treated as sufficient support for California solicitation language.

Allowed human dispositions: accept, accept with modified conditions, reject, defer, or escalate.

Audit boundary: AI advice and human rationale are recorded separately.



07.

Key technical decisions and tradeoffs

Design choices reflect both small-firm constraints and procurement risk.

01

Configuration-first portability

Adapts to ICM vs another profile without code changes.

02

Human-confirmed consequential inputs

Avoids pretending the prototype fully extracts every solicitation fact.

03

Controlled reason codes

Improves auditability and regression testing.

04

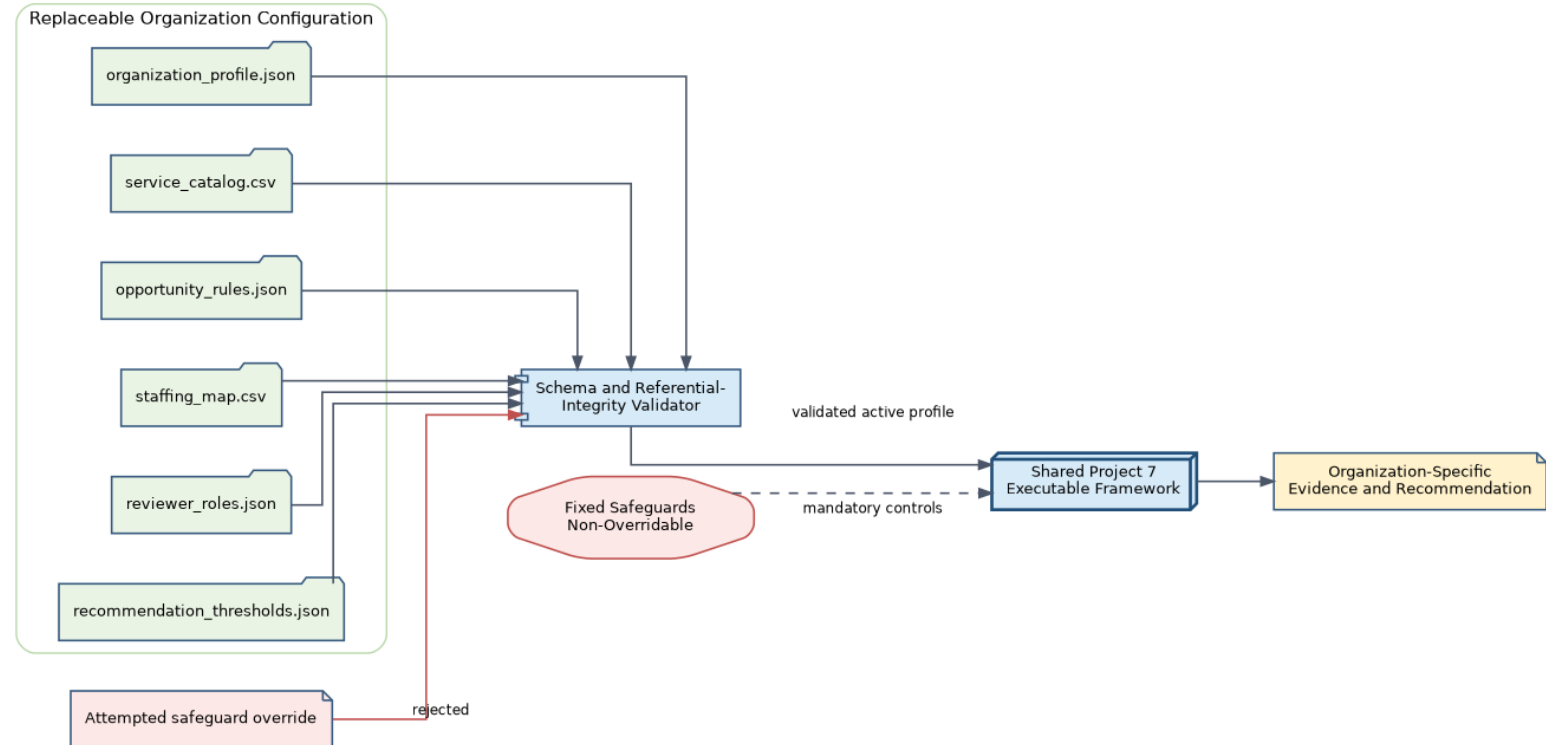
Fail-closed evidence handling

Unsupported conclusions escalate or abstain.

05

Frozen technical candidate

Paper/publication changes do not alter runtime evidence.

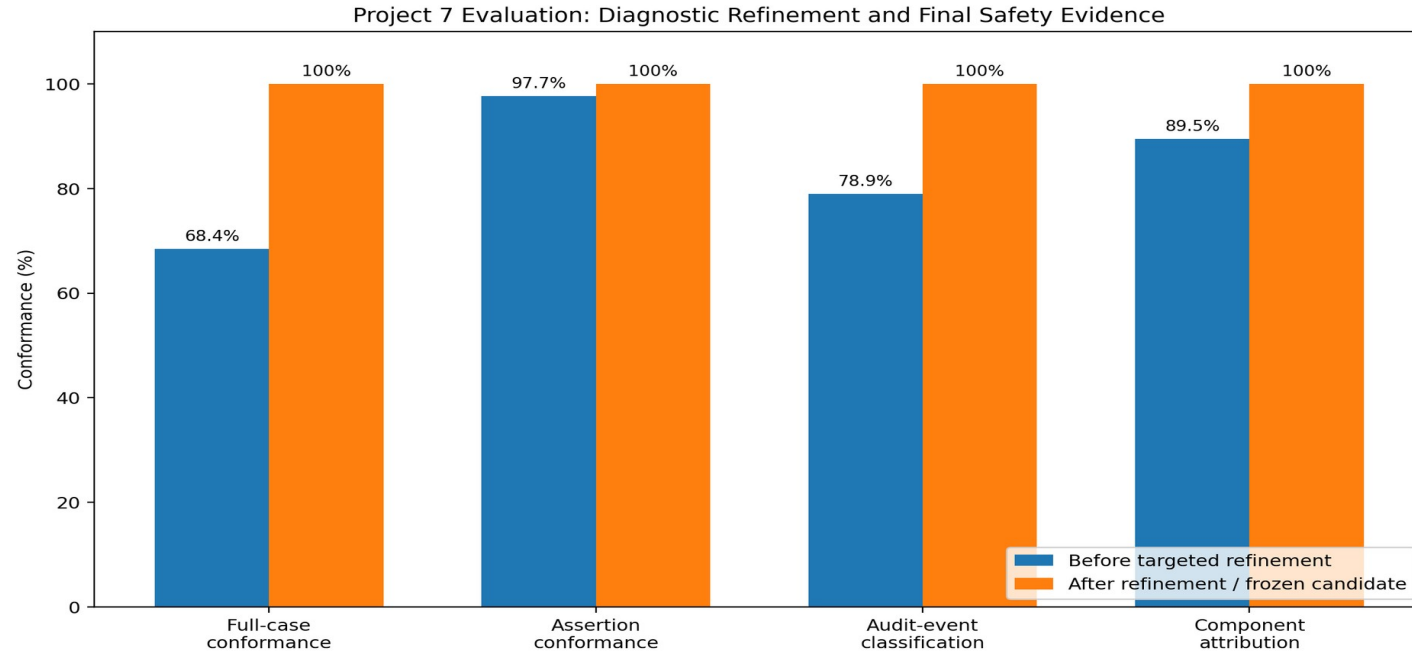


Tradeoff: the prototype favors explainability, traceability, and bounded authority over end-to-end automation.

08.

Evaluation evidence and refinement results

The scenario suite tested behavior, not only isolated components.



Final candidate: 19/19 cases • 262/262 assertions • 167/167 critical assertions • escalation recall 3/3 • fail-closed recall 6/6 • 0 autonomous external actions

19/19

cases executed

262/262

final assertions passing

0

critical or major
unresolved acceptance
defects

6 → 0

audit/attribution
mismatches after
refinement

Interpretation: refinements corrected observability/governance defects while preserving the original scenario boundaries and human-safety behavior.

09.

Responsible AI and governance safeguards

The control design is system-specific: evidence, escalation, audit, and human review.

High-Residual-Risk Governance Matrix: Controls Preserve Human Authority

Risk	High Residual Risk	Prototype Safeguards / Controls	Required Human Authority
GR-06	Public-sector model validity / domain shift	Domain-shift warning; bounded triage; confidence cannot establish legal correctness	Contracts / Legal reviewer
GR-07	Operator-selected passage completeness	Operator confirmation; explicit incomplete-document warning; escalation for ambiguity	Opportunity / Contracts reviewer
GR-08	Bounded authoritative evidence corpus	Citation validation; evidence-sufficiency gate; abstain/escalate when support is inadequate	Qualified specialist
GR-19	Records retention and legal hold	Versioned audit/evidence records; production gate remains unsatisfied	Records / Legal authority
GR-20	Production security, IAM, and operations	No production-ready claim; no external writes; deployment gates and independent approval	Security + Executive authority

Transparency:

- intermediate outputs, limitations, evidence status, and reason codes are visible

Accountability:

- case IDs, packet IDs, audit events, and human rationale are preserved.

Human oversight:

- recommendations identify reviewer roles and allowed dispositions.

Bounded autonomy:

- no autonomous legal, contractual, or pursuit commitment.

10.

Limitations, risks, and production boundaries

A credible prototype must state where it stops.

Scope

Operator-confirmed metadata and selected clause passages; no complete PDF semantic extraction claim.

Evidence

FAR examples do not substitute for solicitation-specific authority.

Availability

Prototype does not yet demonstrate enterprise SLA, backup, or disaster-recovery controls.

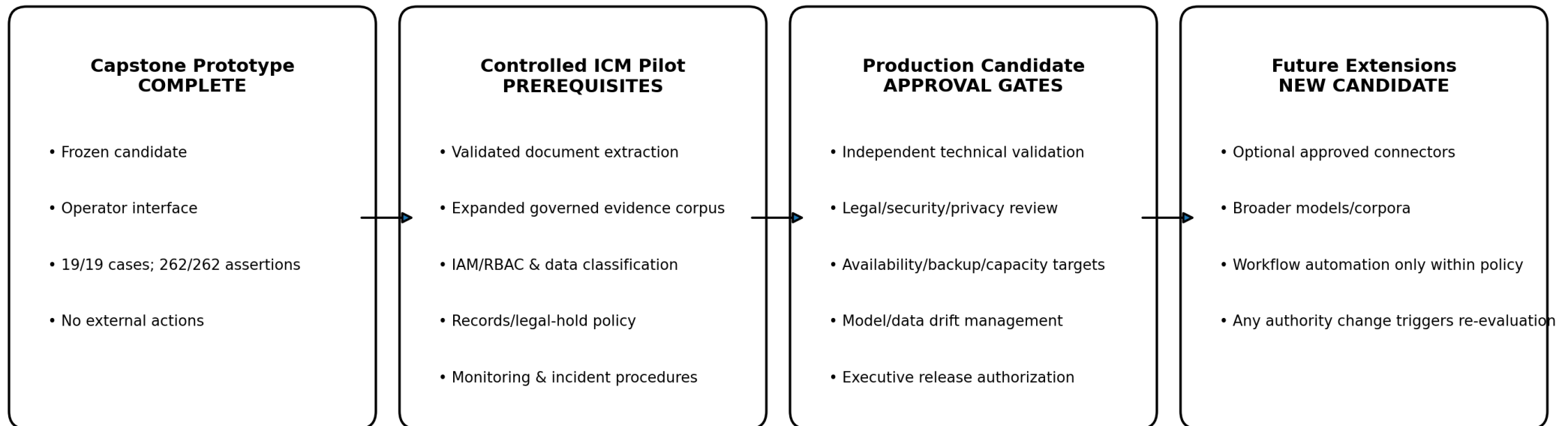
Domain shift

CUAD-trained clause triage is not public-sector legal certainty.

Operations

IAM/RBAC, records retention, legal hold, monitoring, and incident response remain production gates.

Phased Implementation Roadmap from Capstone Prototype to Governed Production



Boundary statement: Project 7 is a governed capstone prototype, not a production-ready procurement or legal authority.

11.

Professional and industry relevance

The value is cross-domain fluency translated into a governed business workflow.

System-level architecture: data, models, orchestration, safeguards, and operator workflow work together.

Cross-domain synthesis: statistical context + deep learning triage + evidence workflow + governance.

Small-business relevance: improves pursuit discipline while preserving quality control and specialist judgment.

Portfolio value: shows ability to defend tradeoffs, not just produce artifacts.

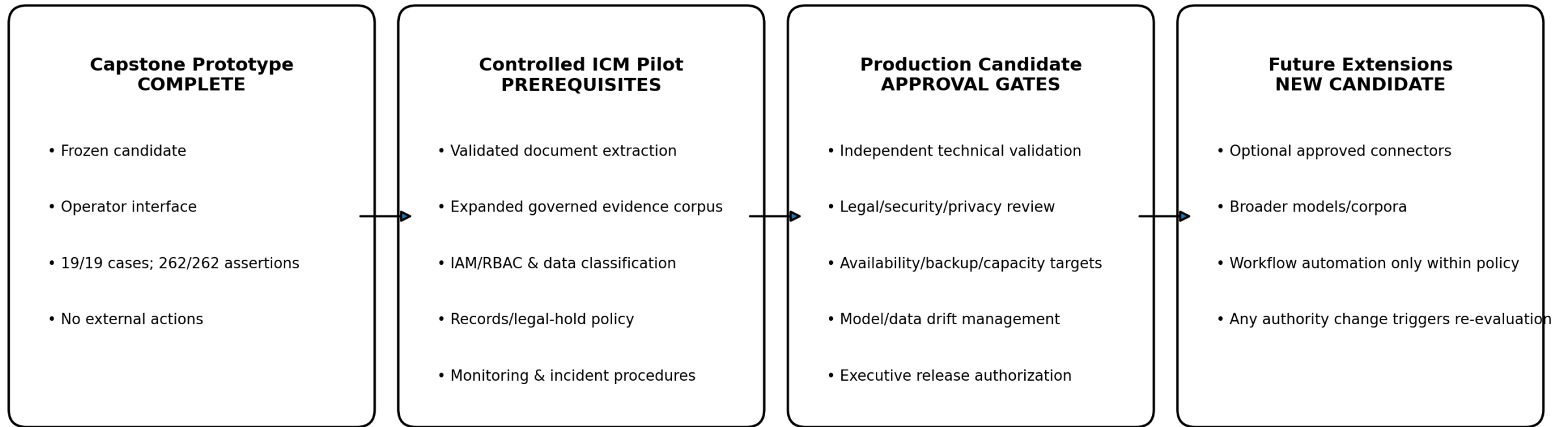
The final product is a reviewer-facing decision-support workflow aligned to ICM's real operating context.



Next steps and future improvements

Future work proceeds through production gates rather than feature accumulation.

Phased Implementation Roadmap from Capstone Prototype to Governed Production



Progression is gated by objective evidence and named human acceptance authority; the capstone does not claim production readiness.

Validate full-document extraction and source authority coverage.

Expand scenario testing and edge-case stress evaluation.

Implement IAM/RBAC, retention, legal-hold, monitoring, and incident-response controls.

- Calibrate reviewer workload, thresholds, and business acceptance criteria with domain experts.

Defense close

Configurable Opportunity-to-Contract Intelligence & Assurance Framework

Core defense statement

This project demonstrates capstone-level synthesis by integrating prior analytical, modeling, and agentic methods into a bounded, auditable, human-governed workflow for public-sector consulting decisions.

Thank you.



Appendix A — Artifact and rubric crosswalk

Backup slide for mentor questions about evidence location and rubric coverage.

Rubric expectation	Where shown	Evidence / artifact
Integrated system	Slides 4–7	Operator interface, architecture, workflow, packet outputs
Cross-domain integration	Slide 5	Projects 1/2/4/6 runtime; Projects 3/5 bounded design evidence
Evaluation outcomes	Slide 9	19 cases, 262 assertions, refinement metrics, CI evidence
Ethics and safeguards	Slides 7, 10–11	Nonbinding recommendations, evidence sufficiency, audit, human disposition
Professional readiness	Slides 12–13	ICM use case, production gates, portfolio-level synthesis
Reproducibility	Repository	Frozen candidate, overlay manifests, Quality Gate #28, final paper